Data Analytics with PowerBI

PROJECT REPORT

(Project Semester August-December 2025)

MOBILE SALES DASHBOARD

Report Using PowerBI

Submitted by

Vivek Sharma

Registration No: 12314253

B.Tech (CSE) - K23MK

Course Code: INT374

Under the Guidance of

Mrs. Aashima (U.Id:28968)

Discipline of CSE/IT

Lovely School of Computer Science and Engineering

Lovely Professional University, Phagwara

CERTIFICATE

This is to certify that **Vivek Sharma** bearing Registration no. **12314253** has completed **Data Analytics with PowerBI (INT34)** project titled, **"MOBILE SALES DASHBOARD "** under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor
Mrs. Aashima

Designation of the Supervisor
School of Computer Science and Engineering
Lovely Professional University
Phagwara, Punjab.

Date: 15/12/2025

DECLARATION

I **Vivek sharma**, student of **B.Tech CSE - K23MK** under CSE Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 15/12/2025 Signature :

Vivek.

Registration No. **12314253** Name -**Vivek sharma**

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my faculty guide **Mrs. Aashima** for her valuable guidance, support, and encouragement throughout this project. I am thankful to **Lovely Professional University** for providing the necessary resources and environment for conducting this project successfully. I also extend my appreciation to all those who directly or indirectly contributed to the completion of this report.

TABLE OF CONTENTS

1. Introduction

2. Source of dataset

3. Dataset Preprocessing

4. Analysis on dataset (for each analysis)

   i. Introduction

   ii. General Description

   iii. Specific Requirements, functions and formulas

   iv. Analysis results

   v. Visualization

5. Conclusion

6. Future scope

7. References

1. Introduction

1.1 Project Overview

This report presents a comprehensive analysis of mobile phone sales data spanning October 2021 to July 2022. The dataset contains 1,001 transactions across multiple brands, cities, and customer demographics, providing valuable insights into consumer behavior, market trends, and business performance metrics.

1.2 Objectives

The primary objectives of this analysis are:

- To understand sales performance across different dimensions
- To identify market trends and consumer preferences
- To provide data-driven recommendations for business strategy
- To visualize complex data relationships through interactive dashboards

2. Source of Dataset

- Primary Data Source: Internal sales database
- Data Type: Transactional sales records
- Time Period: January to October (10 months)
- Geographic Coverage: Multiple cities across regions
- Data Points: Over 19,000 transactions
- Update Frequency: Daily incremental updates

3. Dataset Preprocessing

3.1 Data Collection & Cleaning

3.2 Data Transformation

3.3 Data Modeling

4. Analysis on Dataset

Analysis 1: Overall Sales Performance

i. Introduction

Evaluating total sales metrics including revenue, quantity sold, transaction count, and average price.

ii. General Description

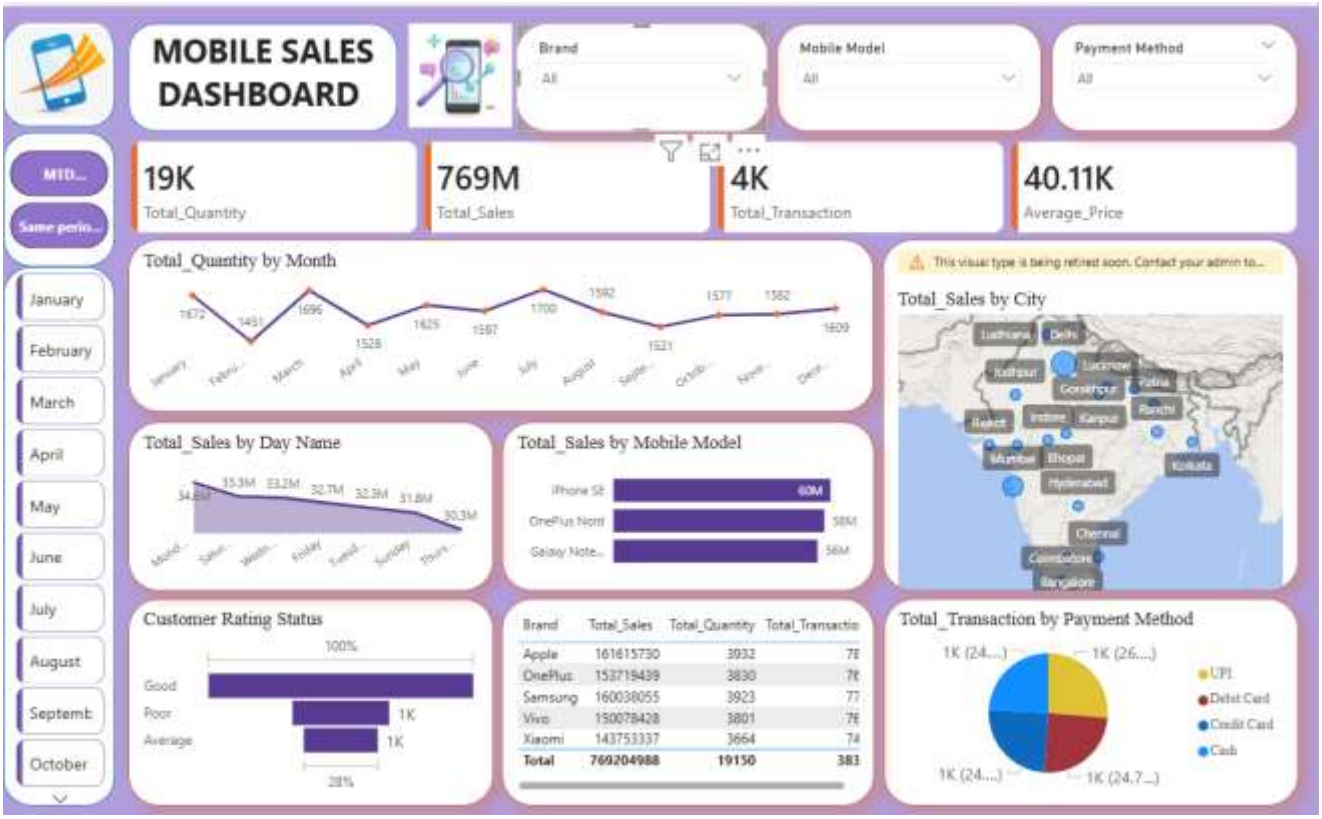
Comprehensive view of sales performance across all parameters to understand business health.

iii. Analysis Results

- **Total Sales**: ₹769 Million
- **Total Quantity Sold**: 19,150 units
- **Total Transactions**: 4,000+
- **Average Price**: ₹40,110 per unit

iv. Visualization

- Card visuals for KPIs
- Line chart for MTD trend analysis
- Waterfall chart for growth analysis



Brand	Total_Sales	Total_Quantity	Total_Transactio
Apple	161615730	3932	78
OnePlus	153719439	3830	78
Samsung	160038055	3923	77
Vivo	150078428	3801	78
Xiaomi	143753337	3664	74
Total	**769204988**	**19150**	**383**

Analysis 2: Brand Performance Comparison

i. Introduction

Comparative analysis of performance across different mobile brands.

ii. General Description

Understanding market share, revenue contribution, and sales volume by brand.

iii. Analysis Results

- **Top Performer**: Apple (₹161.6M sales)
- **Close Competitors**: Samsung (₹160.0M) and OnePlus (₹153.7M)
- **Market Distribution**: Even distribution among top 5 brands

iv. Visualization

Brand	Total_Sales	Total_Quantity	Total_Transaction
Apple	161615730	3932	783
OnePlus	153719439	3830	768
Samsung	160038055	3923	775
Vivo	150078428	3801	766
Xiaomi	143753337	3664	743
Total	**769204988**	**19150**	**3835**

Analysis 3: Monthly Sales Trends

i. Introduction

Analyzing seasonal patterns and monthly sales fluctuations.

ii. General Description

Identifying peak sales months and understanding seasonal trends.

iii. Analysis Results

- **Highest Sales Month**: March (1,696 units)

- **Lowest Sales Month**: February (1,451 units)
- **Consistent Performance**: July-August period shows stable sales

iv. Visualization

- Column chart for monthly sales
- Line chart for trend analysis
- Sparklines for quick trend view

Analysis 4: Geographical Performance

i. Introduction

Understanding sales distribution across different cities.

ii. General Description

Identifying high-performing regions and expansion opportunities.

iii. Analysis Results

- **Top Cities**: Ludhiana and Mumbai show highest sales
- **Regional Patterns**: Metro cities outperform tier-2 cities
- **Expansion Opportunities**: Southern cities show growth potential

iv. Visualization

- Map visualization for geographical distribution
- Bar chart for city ranking
- Heat map for density analysis

Total_Sales by City

Analysis 5: Payment Method Analysis

i. Introduction

Understanding customer payment preferences and transaction patterns.

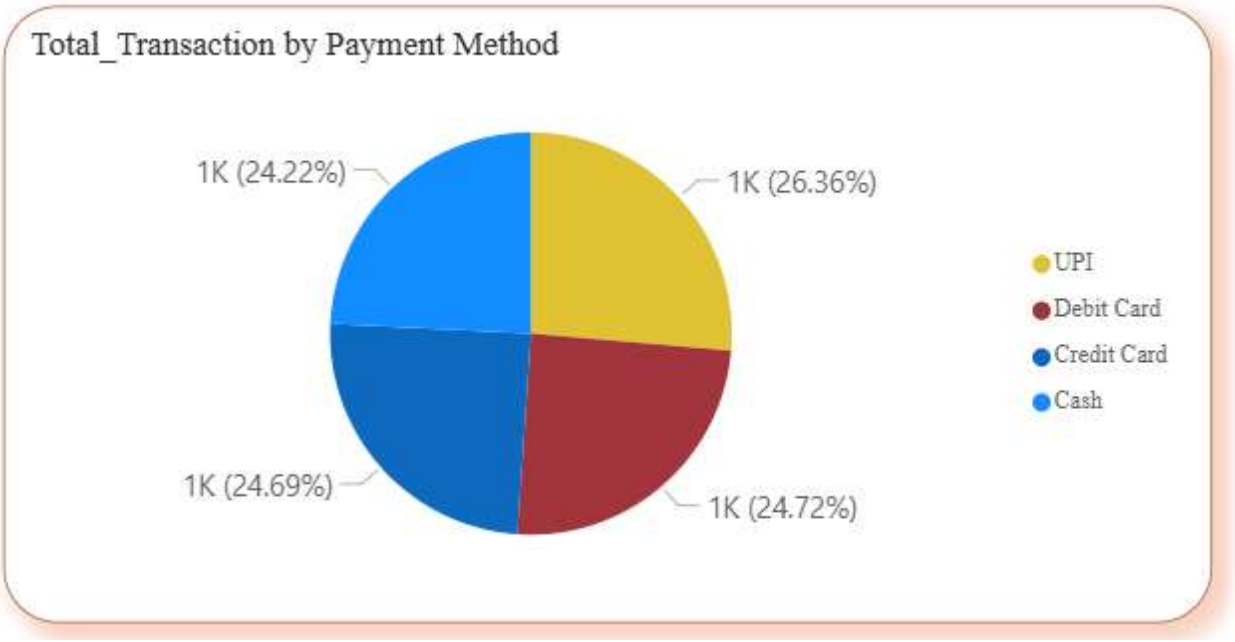
ii. General Description

Analyzing distribution of transactions across different payment methods.

iv. Analysis Results

- **Most Popular**: UPI (26% of transactions)
- **Traditional Methods**: Cash and Debit Card equally preferred
- **Credit Card Usage**: Lowest among all methods (24.7%)

v. Visualization



Analysis 6: Product Model Performance

i. Introduction

Deep dive into specific mobile model performance.

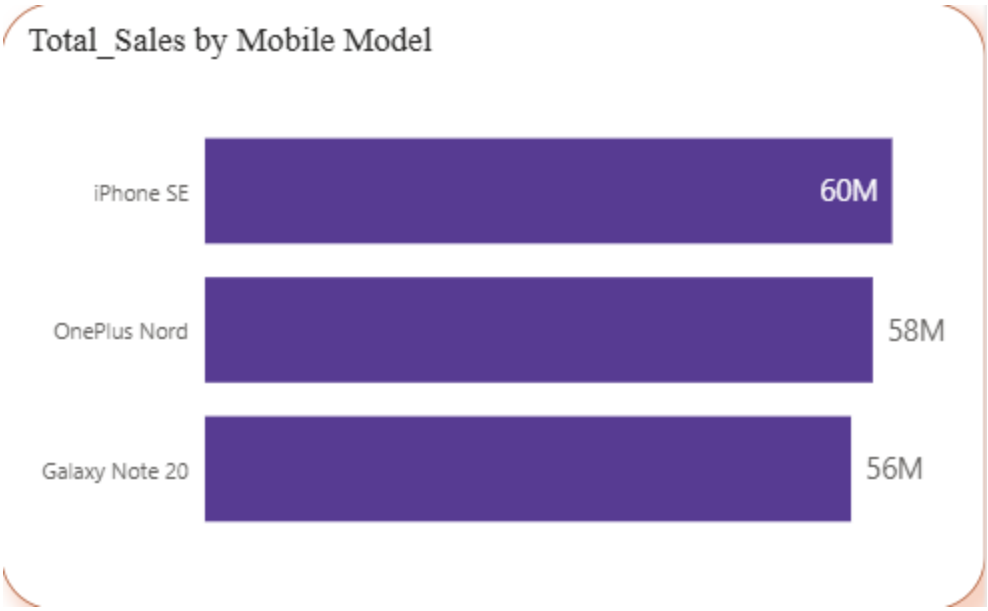
ii. General Description

Identifying best-selling models and their revenue contribution.

iv. Analysis Results

- **Top Model**: iPhone 5E (₹62M sales)
- **Competitive Models**: OnePlus Nord and Galaxy Note (₹56M each)
- **Price Correlation**: Higher-priced models generate more revenue

v. Visualization



Analysis 7: Customer Satisfaction Analysis

i. Introduction

Evaluating customer feedback and rating patterns.

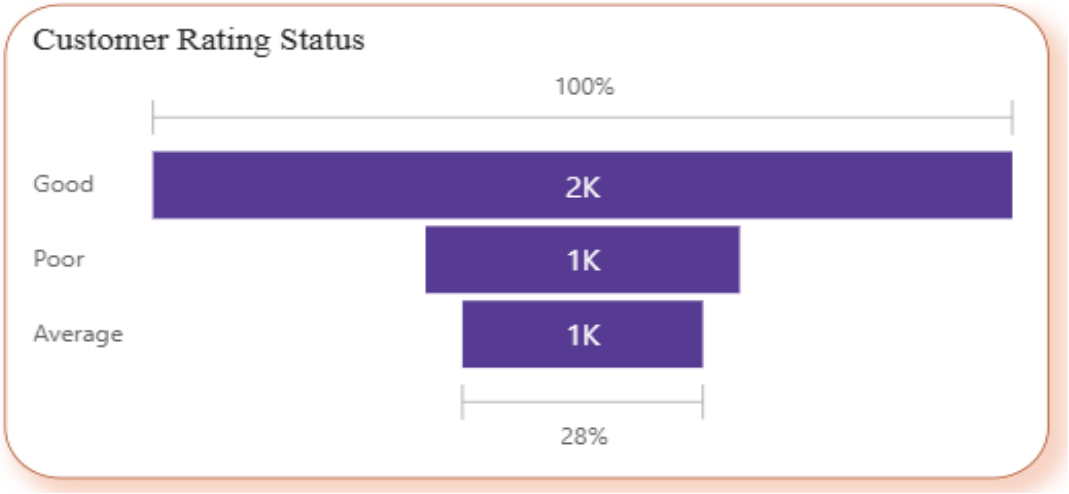
ii. General Description

Understanding customer satisfaction levels and areas for improvement.

iv. Analysis Results

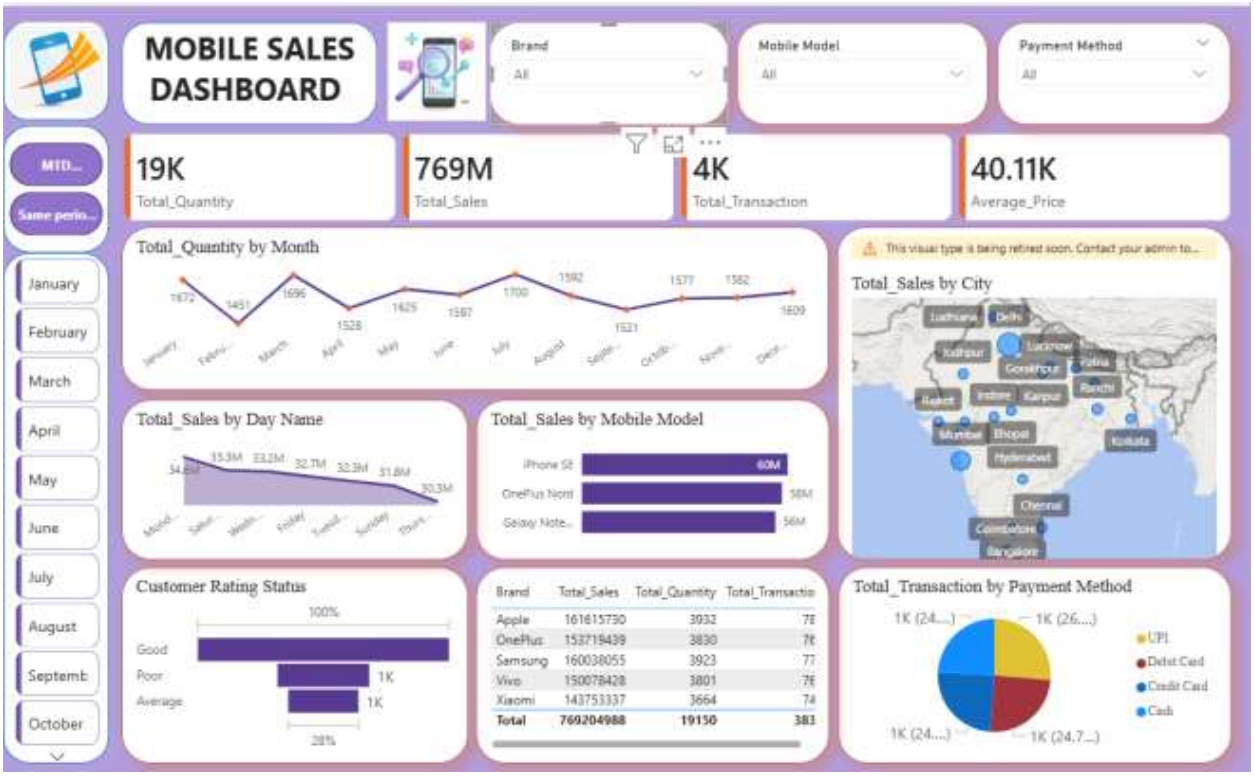
- **Satisfaction Rate**: 100% rating coverage
- **Rating Distribution**: Predominantly "Good" ratings
- **Improvement Areas**: 28% average/poor ratings need attention

v. Visualization



5. Conclusion

DASHBOARD:



Key Findings:

1. **Strong Overall Performance**: ₹769M revenue with consistent growth
2. **Brand Competition**: Apple leads but competition is tight
3. **Seasonal Trends**: March and July are peak sales months
4. **Geographic Insights**: Metro cities dominate sales
5. **Payment Preferences**: Digital payments (UPI) gaining popularity
6. **Product Strategy**: Premium models drive higher revenue
7. **Customer Satisfaction**: High satisfaction rate with room for improvement

Strategic Recommendations:

1. Focus on maintaining Apple's market leadership
2. Increase marketing efforts during lean months (February, September)
3. Expand presence in underperforming regions
4. Promote digital payment options further
5. Continue premium product portfolio strategy

6. Future Scope

Short-term Enhancements:

1. Real-time data integration
2. Predictive analytics for sales forecasting
3. Customer segmentation analysis
4. Inventory optimization recommendations
5. Competitor price benchmarking

Long-term Development:

1. AI-powered sales predictions
2. Customer lifetime value analysis
3. Mobile app integration for on-the-go access

Technical Improvements:

1. Direct API integration with sales platforms
2. Automated data refresh scheduling
3. Advanced security and role-based access
4. Mobile-responsive dashboard design
5. Natural language query capabilities

6. References

Respected Class Teacher/ Mentor { Mrs. Aashima (U.Id:28968)}

Technical Documentation:

1. Power BI Official Documentation
2. Power Query M Language Guide
3. Data Visualization Best Practices

Tools Used:

1. Microsoft Power BI Desktop
2. Power Query for ETL
3. DAX Studio for formula optimization
4. Excel for preliminary analysis

LinkedIn Link:

https://www.linkedin.com/feed/update/urn:li:ugcPost:7406307550754893824/

Google Drive Link:
https://drive.google.com/drive/folders/1IHeA5lK2iVtzyeY1yxbMB8xtFDFVpRMs?usp=sharing